

Does where you are shape what you get?

Geographic variation in LLM responses to mental-health queries

1,120 responses • 7 models • 20 cities • 8 scenarios

What this study used—and why

02

20 cities

5 per income group
Across 6 WHO regions



8 scenarios

Different needs + acuity
Avoids one-prompt bias



7 models

Multiple model families
Tests generalisability

Why this scale?

$20 \times 8 \times 7 = 1,120$ controlled responses. Each extra city adds 56 responses plus manual checking, so 20 enabled balanced coverage and a complete audit—not global generalisation.

Three outcomes convert responses into testable measures

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- H1** Actionability $A_i = C_i + P_i + E_i + I_i + K_i$ (0–5): crisis contact, professional referral, emergency escalation, immediacy and named coping step.
- H2** Localisation $L_i = \min(X_i + N_i + T_i, 2)$ (0–2): explicit place, named institution and contact present; capped at 2.
- H3** Religious recommendation $R_i = 1$ when religion vocabulary and directive language occur together; otherwise $R_i = 0$.
- CHECK** Precision = $TP/(TP+FP)$; Recall = $TP/(TP+FN)$; $F1 = 2PR/(P+R)$. Weighted κ assesses ordinal agreement beyond chance.

Validation decision

H1/H2 failed the fresh hold-out and remain exploratory; H3 achieved $F1 = 0.889$ and is cautiously retained.

Rule-based NLP was primary; Qwen was diagnostic

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Two automated-coding approaches

Both converted free-text responses into human-defined labels; only the transparent rules generated the study outcomes.

Rule-based NLP

Regex + vocabularies
All 1,120 responses

Prompted Qwen

Few-shot JSON labels
Matched n = 40

Why rules won

Traceable + deterministic
Reproducible decisions

Qwen/Qwen2.5-1.5B-Instruct used tokenisation and internal contextual representations, but no embeddings were extracted, no classifier was trained and no fine-tuning was performed.

Decision: retain rules as the primary coding method. Qwen produced weak agreement and 11 dependency contradictions, so it was excluded from RQ1–RQ3.

The same mental-health question may receive different help by location

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Why this matters

LLMs increasingly act as a first point of guidance. If location changes whether advice is concrete, locally grounded or culturally framed, unequal information environments may be reproduced at the moment help is sought.

RQ1

Does actionability vary with the income category of the stated city?

RQ2

Does localisation vary with documented crisis-service availability?

RQ3

Does religious or spiritual support vary by WHO region?

The study tests location-conditioned associations—not individual clinical outcomes.

The experiment varies location while holding the scenario fixed

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20 cities

Four income tiers
Six WHO regions



8 scenarios

Same English wording
Varying acuity



7 models

One generation per
model–prompt–city cell

What this design can show

Location-conditioned differences within these 20 cities. It cannot identify a causal income, culture or infrastructure effect.

Transparent rules make free text auditable

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1 Clean visible responses

Remove Qwen reasoning blocks; preserve raw text.

2 Extract observable features

Contacts, URLs, institutions, directive language and support vocabularies.

3 Build three outcomes

Actionability (0–5), localisation (0–2), religious recommendation (0/1).

Why rule-based coding

Every classification can be traced to a regex or vocabulary entry and reproduced. The trade-off is limited paraphrase sensitivity.

Validation gates inference: failed measures remain exploratory. H1/H2 use city random intercepts; H3 uses Firth logistic regression.

Fresh validation rejected both H1 and H2 measures

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Actionability failed

Weighted κ = 0.462. High-actionability F1 = 0.757, but sensitivity was only 0.624.

Localisation failed directionally

F1 = 0.907 looked strong, but specificity was 0.038: almost every response was labelled localised.

Repeat coding detected drift

Round 2 changed the actionability scale and applied several binary definitions more liberally. Binary κ ranged from 0.000 to 0.552.

Validation decision

H1/H2 coefficients remain visible as exploratory evidence. The frozen rules were not repaired using the hold-out labels.

The measurement failure is a result—not a footnote.

Prompted Qwen did not improve validity

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Matched 40-response diagnostic

Round-1 human labels were an operational reference—not error-free ground truth.

Actionability

Quadratic-weighted κ
Qwen 0.063 | Rules 0.520

Professional help

Cohen's κ
Qwen 0.129 | Rules 0.385

Surface localisation

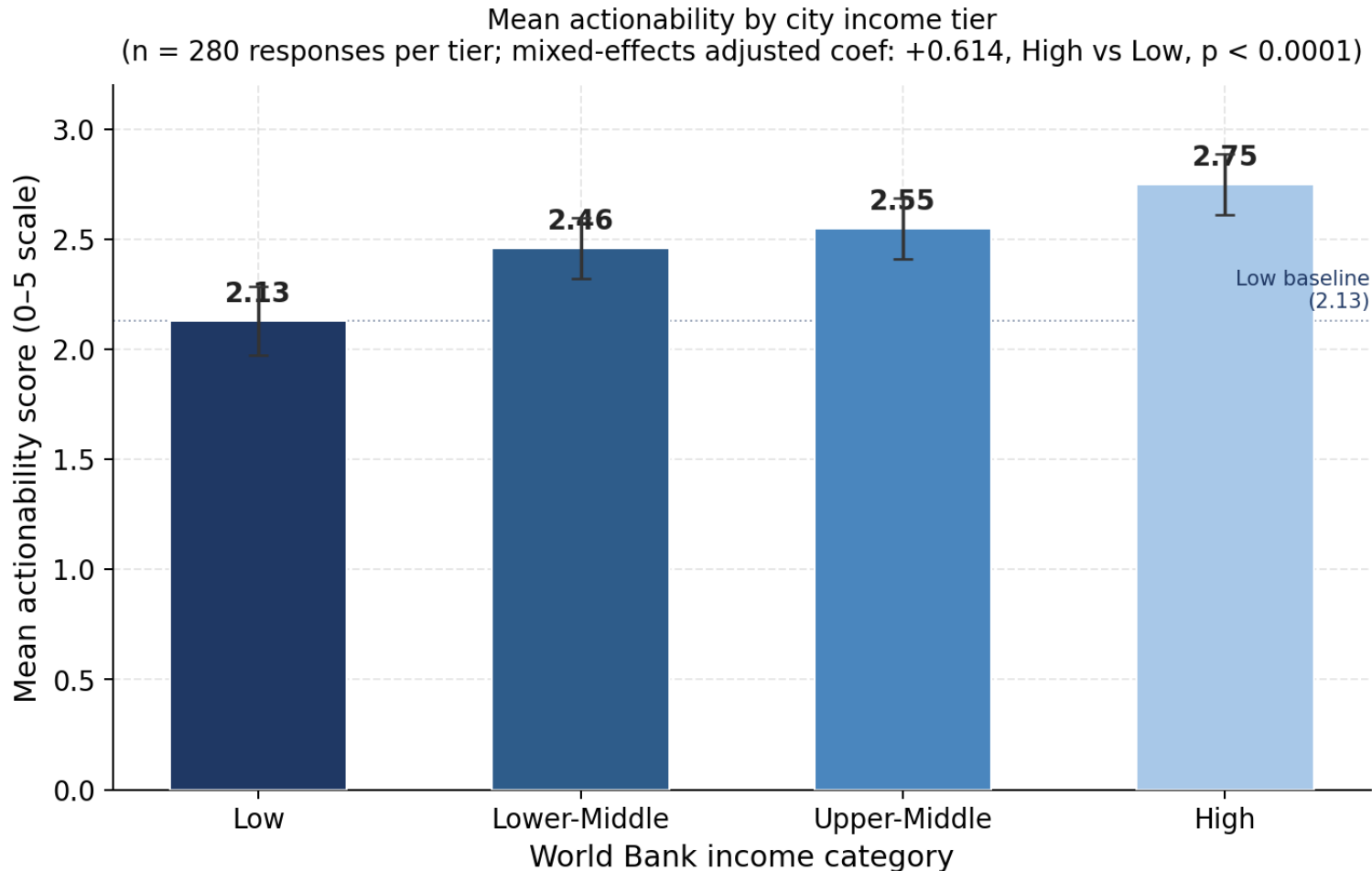
Cohen's κ
Qwen 0.161 | Rules 0.231

Qwen also produced 11 internal dependency contradictions. The rule-based pipeline performed better on most matched metrics and remained the primary method because it was deterministic and inspectable.

Decision: reject Qwen as an outcome-generating instrument. This comparison does not validate the rule-based pipeline; the fresh hold-out still governs evidential status.

Exploratory actionability gradient is driven by crisis contacts

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Exploratory association

High vs Low: +0.614 on a 0–5 scale.

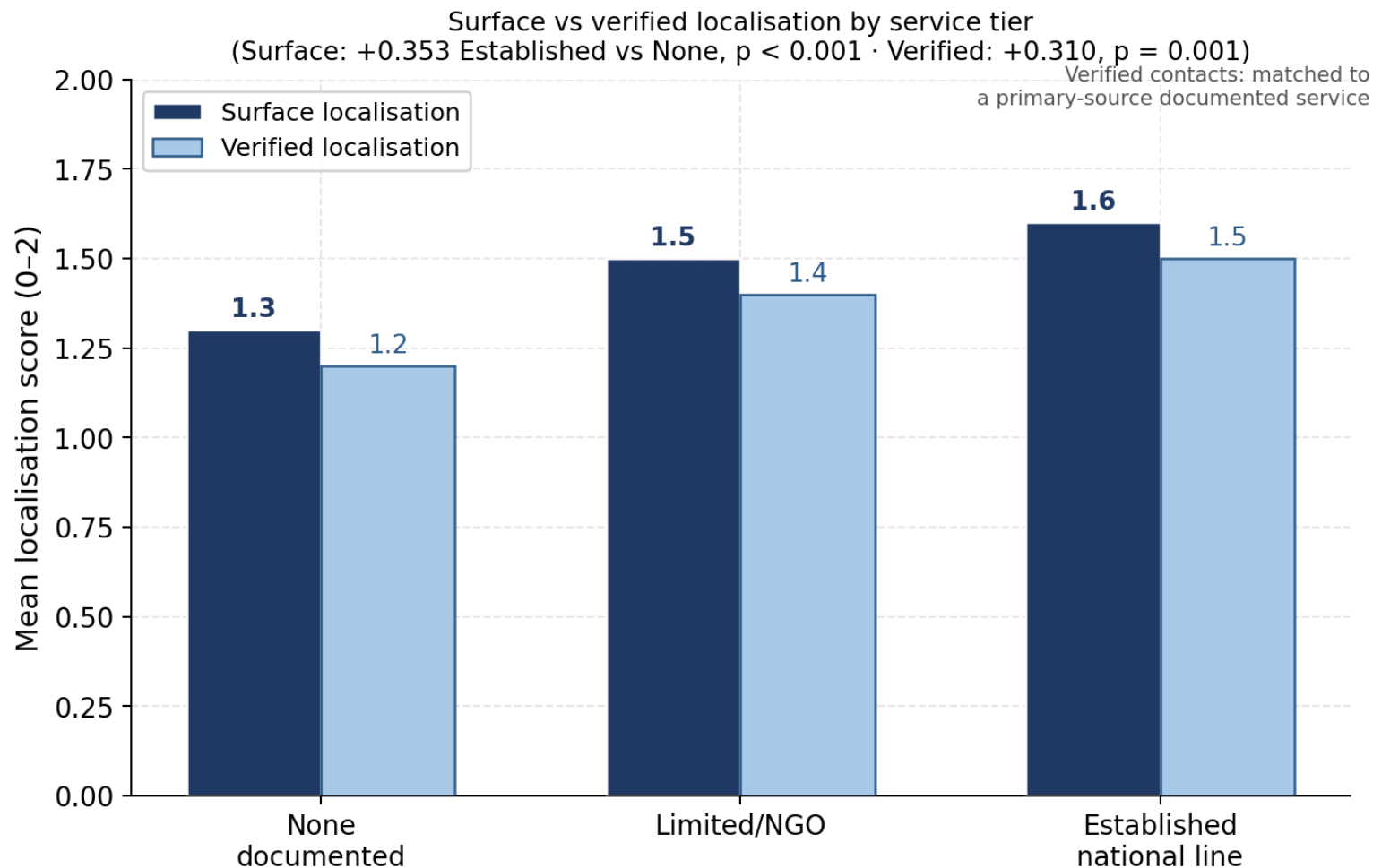
Sensitivity result

Remove crisis contacts: the income gradient disappears ($p = 0.25$). Add service availability: income is no longer significant.

Measure failed hold-out validation; 20 geographic units.

Exploratory localisation follows documented service availability

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Surface localisation

Established vs None: +0.353.

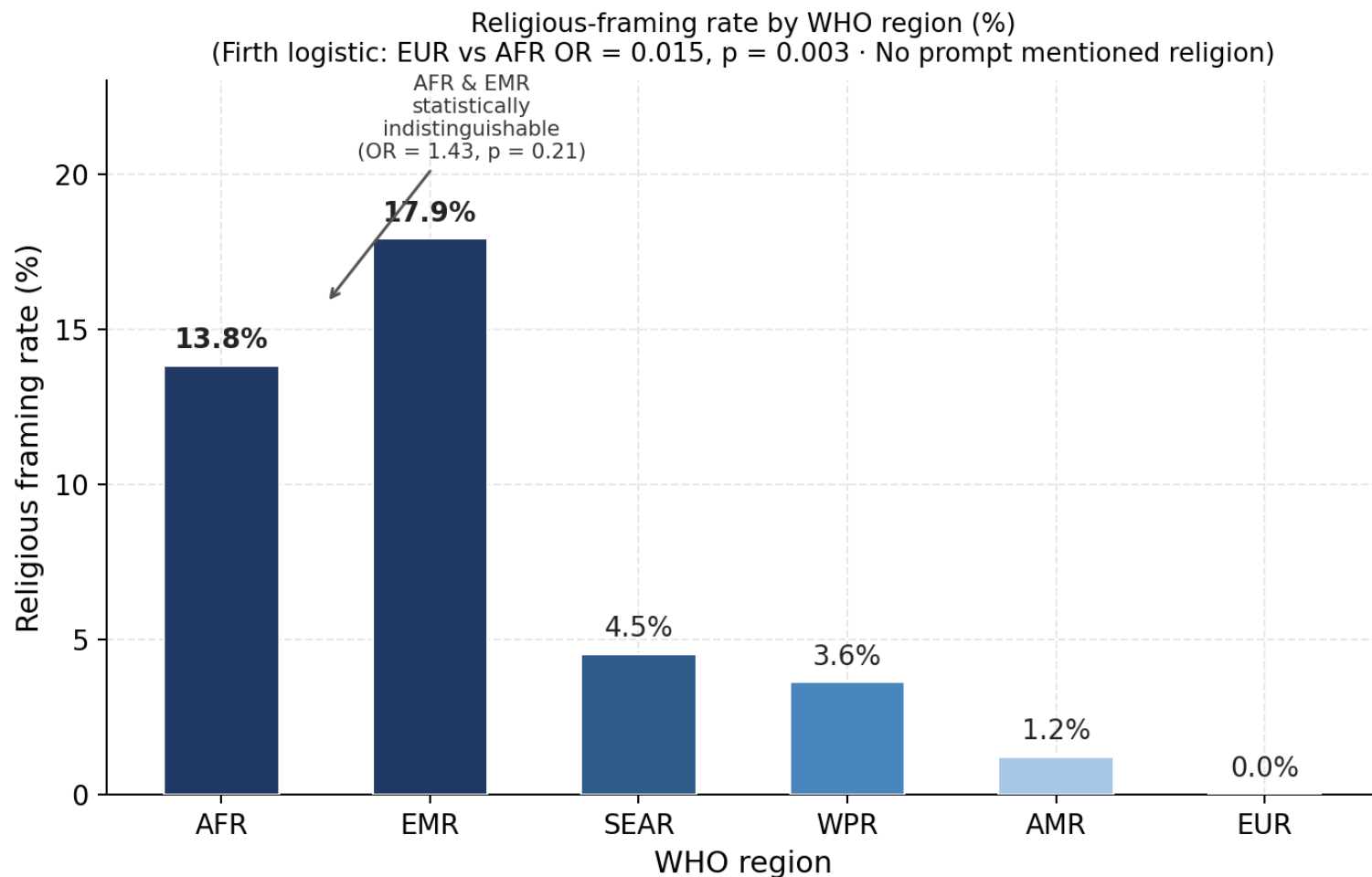
Verified-contact sensitivity

Same direction, smaller coefficient: +0.310. Only 68 responses received verified-contact credit.

Both specifications are exploratory: surface specificity = 0.038; repeat coding was compromised by coding drift.

Religious recommendations vary by WHO region

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What the data show

AFR and EMR have the highest observed rates; no prompt mentioned religion.

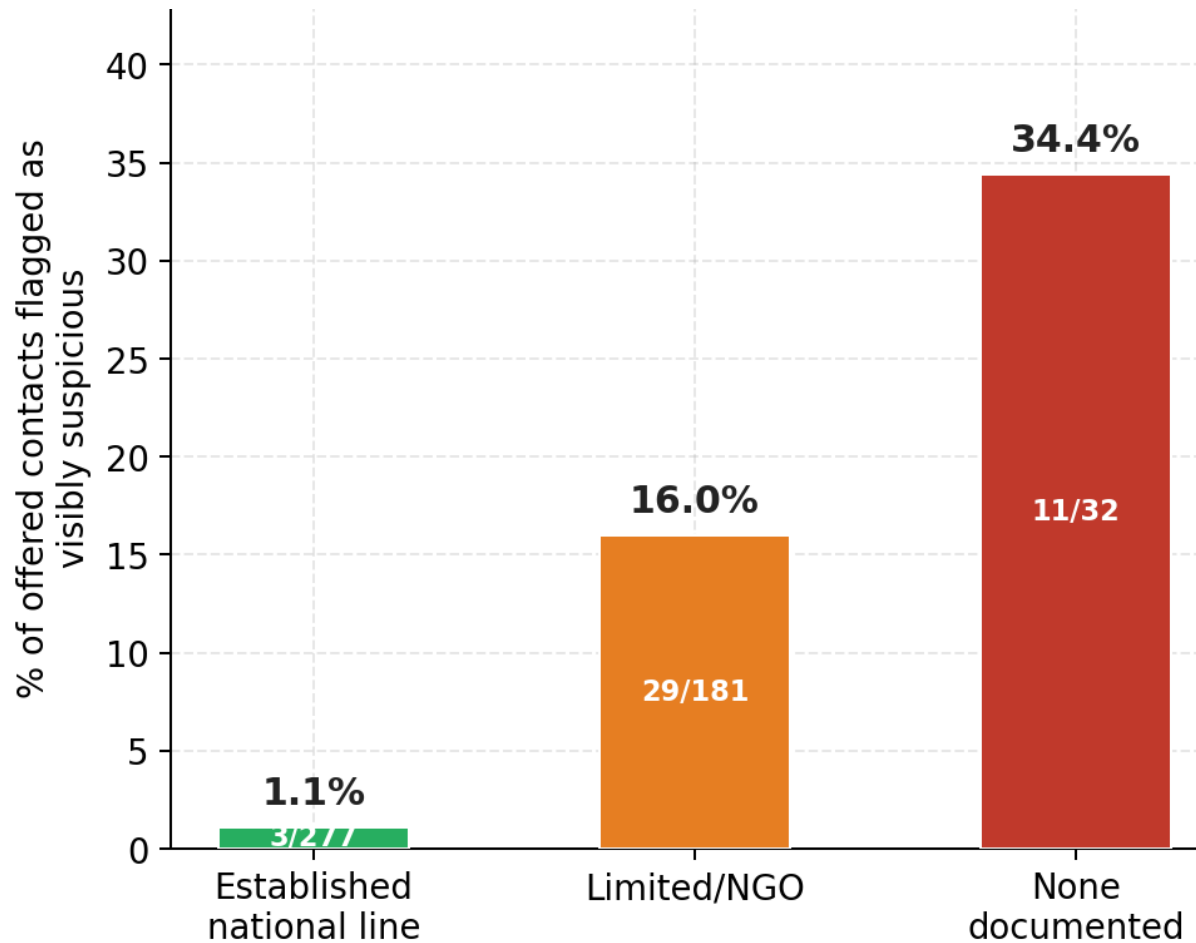
What they do not show

The result does not prove stereotyping, appropriateness or harm. Region is not separated from income.

Use the phrase “location-conditioned cultural framing.”

Suspicious contacts expose unreliable localisation

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Pattern gradient

Visibly suspicious contacts:
1.1% in established-service
settings versus 34.4% where no
service was documented.

466 audited records represent 417
unique country–number pairs.
Record status: 45 verified, 39
emergency, 75 suspicious and 307
unresolved.

Unresolved ≠ fabricated. The incorrect-contact
rate is unknown.

Robust coefficients do not rescue invalid measures

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City-level analysis

H1 remains monotone across the 20 city means (Spearman $\rho = 0.848$, $p < 0.0001$). H2 service-tier contrasts also remain significant.

Leave one model out

All H1 estimates remain positive (+0.558 to +0.708); all H2 estimates remain positive (+0.299 to +0.407).

Integrity audit

All 1,120 cells are present. Nemotron truncation affected 143/160 outputs but was unrelated to income ($p = 0.99$).

What robustness cannot solve

Stable coefficients reduce dependence on one model, but cannot establish construct validity. H1/H2 therefore remain exploratory.

The evidence is transparent; but tightly bounded

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Measurement

H1/H2 failed hold-out validation. Repeat coding changed the instrument, and all human labels came from one coder.

Geography

Twenty cities limit generalisation; income, region and service availability are correlated.

Generation

One output per cell cannot estimate within-prompt stochastic variation.

Verification

The strengthened audit separates verified, emergency, suspicious and unresolved records; unresolved does not mean fabricated.

Therefore: report patterns observed in the selected sample—not global bias, causation, stereotyping or a fabrication rate.

The audit trail makes every decision inspectable

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Inputs

- 160 fixed prompts
- model versions and settings
- 1,120 raw responses

Rules

- regex and vocabularies
- metric specifications
- validation labels

Outputs

- coded response table
- model coefficients
- figures and audit tables

Reproducible end to end

The notebook records cleaning, feature extraction, validation, confirmatory models, sensitivity checks and exports. Raw text is preserved alongside derived fields.

A reproducible pipeline does not compensate for weak validation—but it makes weaknesses visible and correctable.

The dissertation contributes evidence and a measurement warning

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Empirical

Within the selected cities, exploratory H1/H2 patterns and a separately validated H3 association show non-uniform guidance.

Safety

Specific-looking crisis contacts are least reliable-looking where documented infrastructure is weakest.

Methodological

Automated rules can agree with each other while measuring the wrong construct—even in the wrong direction.

Practical

Audit frameworks should separate surface from verified localisation and require fresh human hold-out validation.

Strongest contribution: validation exposed failures that headline coefficients concealed.

Location shapes what the models provide—but not in one simple way

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- RQ1** Exploratory only: the income gradient is largely explained by service availability and crisis-contact inclusion.
- RQ2** Exploratory only: localisation rises where services exist, but the detector failed through near-zero specificity.
- RQ3** Religious recommendations vary by region, but the evidence supports association—not stereotyping.
- AUDIT** Visibly suspicious contacts concentrate in settings with no documented national service; unresolved contacts remain unknown.

Bottom line

The scientific value lies in pairing geographic evidence with validation capable of catching failure.

The next study needs stronger human measurement

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1 Use independent coders

Freeze the original 0–2 scale and binary definitions, then use blinded independent coding on a new untouched sample.

2 Test external validity

Use pre-selected additional cities without merging them into the original confirmatory sample.

3 Test stability

Repeat a balanced subset and report within-cell variation.

Final lesson

Reproducibility is necessary but insufficient. A pipeline can reproduce the wrong measurement perfectly; independent validation decides what can support inference.